



EDUCATION & SCHOLASTIC ACHIEVEMENTS			
Program		Institution	CGPA / %
BTech, Chemical Engineering		Indian Institute of Technology, Madras	8.88
Class XII (CBSE)		GMSSS-33, Chandigarh	92.4%
Class X (ICSE)		St. Stephen's School, Chandigarh	92.8%
● Ranked in the top 0.13% (99.87 percentile) in JEE Mains 2024 and in the top 0.43% of JEE Advanced 2024 , out of total 1.4 million+ candidates			
INTELLECTUAL PROPERTY AND PUBLICATIONS			
Intellectual Property		Filed a provisional IP for a streaming-naïve multi-agent treasury orchestration for asset allocation and cashflow forecasting	
Publications	1	First author of “An Interactive RL⁶ Environment on ML Experiment Integrity Auditing” paper, under review at NeurIPS 2026	
		Built the first RL environment for frontier LLMs, which surpasses all ML-trained human baselines & achieves 95.4% accuracy	
	2	First author of "When LLMs Orchestrate but Do Not Compute" paper, submitted to SurgeLLM @ ACL 2026, NLP conference	
		Achieved 0% numerical hallucination on audited benchmarks & higher faithfulness than competing financial-agent baselines	
PROFESSIONAL & RESEARCH EXPERIENCE			
RBCCPS² Lab, IISC Research Intern for Robotics x AI Advisor: Shishir Kolathaya May 2026 - Present		Trained and fine-tuned PPO⁸-based quadruped locomotion policies in IsaacLab , and deployed on a real custom quadruped	
		● Co-trained a DreamerV3-style kinodynamic RSSM⁹ world model with PPO generating 5-step CoM & joint-state predictions	
		● Derived a first-principles differential thermal model for 12 gear-ratio-corrected DC motors, eliminating primary motor failure	
		● Validated zero-shot sim-to-real transfer on an 80 kg quadruped: 27° incline & 14 cm stair walking across all types of terrain	
ValenceAI Undergraduate Research in AI for Process Engineering Advisor: S. Abhinav Raman January 2026 - Present		Architected a multi-agent AI system converting natural-language queries into CFD⁷ simulations , flowsheets & deep insights	
		● Developed a model-agnostic intermediate representation & validation stack that separates LLM reasoning from simulation	
		● Implemented graph-augmented retrieval orchestration system achieving 100% execution success across 41 benchmarks	
		● Validated against CFD benchmarks, achieving 0.88% maximum velocity error across diverse geometries and flow regimes	
Neural-lam Open-Source Contributor		Contributed to Neural-LAM's probabilistic forecasting stack through ongoing Graph-EFM integration for weather forecasting	
		● Developed ensemble calibration diagnostics and standardized forecasting interfaces for uncertainty aware forecasting	
PROJECTS			
CLIPSCOPE (Mechanistic Interpretability for Foundation Models) Self Project (May' 26)		Built a mechanistic interpretability pipeline for VLM¹⁴ , discovering latent concepts in CLIP through sparse feature learning	
		● Trained a 6.3M -parameter Sparse Autoencoder on 160K CLIP ¹⁸ ViT-B/16 activations, achieving 89% explained variance	
		● Demonstrated feature steering by intervening on the learned latent directions, producing up to 24.9% shifts in probabilities	
		● Implemented Anthropic-inspired SAE including Top-K sparsity, ghost-gradient feature recovery & decoder normalization	
GRASP (Robotic control) Project Member, iBot Club, CFI, IIT Madras (June '25 - Feb '26)		Built & deployed Vision-Language-Action policies on an actual robotic arm, optimizing for generalizability & interpretability	
		● Engineered in-context learning for real-robot control: formatted teleoperation demonstrations as LLM few-shot prompts	
		● Found & fixed gradient collapse in VLA models using cross-attention pooling, reducing cross-instruction correlation by 85%	
		● Built an open-vocabulary detection + geometric grasp planning pipeline using OpenCV & SayCAN with 90% success in sim	
NEUROSPIKE (Biologically Plausible AI) Project Member, BioTechnology club x Prof Gopalakrishnan Srinivasan, CS department		Demonstrated a biologically plausible alternative to backpropagation, achieving 96.9% accuracy & 0.9999 cosine similarity	
		● Implemented a 2-layer hierarchical Predictive Coding network in Brian2 modelling Free Energy Principle update dynamics	
		● Benchmarked PC network on 5-task split-MNIST ; PC ⁸ -native update rule forgot 68% less than naive backprop finetuning	
		● Demonstrated generative inference (“ dreaming ”) in under 300 steps by clamping output neurons to target MNIST classes	
RELEVANT COURSEWORK & SKILLS			
Multivariate Data Analysis		Machine Learning in Biology	Probability,Statistics & Stochastics
Machine Learning Specialization (stanford)		MATLAB, Simulink and ASPEN	GenAI with LLMs (by AWS)
			Deep-Learning with Computer Vision
Technical Skills		Languages: Python, Java, C++, C#, SQL, HTML, CSS, JS	
		Frameworks: FastAPI, React, Node.js, MATLAB, Aspen, AutoCAD, Unity, Github	
		AI & ML: Python, Pytorch, Pandas, Numpy, Scikit-learn, Matplotlib, vLLMs, TensorRT-LLM, HuggingFace, Locust, SQL, Matlab, C++, Docker	
COMPETITIONS AND TECHNICAL EXPERIENCE			
COMPETITIONS	Pathway 14th Inter IIT Tech Meet, AI contingent member	Represented IIT-Madras in a national competition, achieving 5th rank among all 23 IITs, building a real-time treasury system	
		● Implemented UCT-MCTS pipeline routing for multi-domain LLM orchestration which achieves 0% numerical hallucination	
		● Designed autoregressive CashFlow forecasting with Gaussian HMM regime detection, reducing RMSE by about ~92%	
		● Finetuned DeBERTa for bond query classification & Llama as domain-specific financial guardrail achieving <50ms latency	
iBot club, IITM	OpenEnv Hackathon Meta x PyTorch x HuggingFace, Represented AI club, CFI, IITM	Selected among top teams for the offline grand finale conducted in Bangalore out of 70,000+ registered developers nationally	
		● Built a first-of-its-kind RL ⁶ environment to train LLM project managers under deception & long-horizon software crises	
		● Trained LoRA-GRPO policies on Qwen-1.5B , evaluating strategic cross-verification under partially falsified observations	
		● Engineered adaptive multi-agent dynamics via deceptive LLM teammates, social testimony graphs & crisis escalation	
Part of the Robotics club	Project Lead, Regensis (June 2026 - Current)	Leading the Software Module of Project Regensis, a hybrid EEG-EMG neuroprosthetic for upper-limb prosthesis control	
		● Designed a CNN-based 5-state FSM for real-time motor intent decoding from EEG power envelopes across 16 channels	
		● Built dual NMF synergy pipelines for EMG and EEG, fusing activation matrices into a Hill-MSK model for joint kinematics	
	Coordinator (June 2025 - Feb 2026)	Led the Software Module of the Robotics Club, engaging 700+ students in robotics through sessions, workshops and events	
		● Built AxiomML, a NumPy-based ML library implementing 15+ algorithms and 25+ neural-network components from scratch	
		● Designed and led 45 students in PPO-based Walker2D locomotion, covering reward engineering and policy optimization	

* - ongoing | 1. LLM - Large Language Models | 2. RBCCPS - Robert Bosch Centre for Cyber Physical Systems | 3. DL - Deep Learning | 4. CV - Computer Vision | 5. CNN - Convolutional Neural Networks | 6. RL - Reinforcement Learning | 7. CFD - Computational Fluid Dynamics | 8. PPO - Proximal Proxy Optimisation | 9. RSSM - Recurrent State Space Models | 10. Spiking Neural Networks | 11. LIF - Leaky Integrate-and-Fire | 12. VLA - Vision Language Action | 13. FX - Foreign Exchange | 14. VLM - Vision Language Models | 15. PC - Predictive Coding | 16. NLP - Natural Language | 17. CFI - Center for innovation | 18. CLIP - Contrastive Language-Image Pre-training | 19. CVAE - Conditional Variational Autoencoder